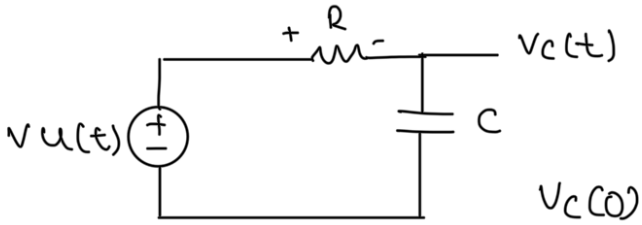
Time domain analysis:



$$RC\frac{dV_c(t)}{dt} + V_c(t) = V\,u(t)$$

$$V_c(0) = 0 \quad t \le 0$$

$$t > 0 \Rightarrow RC\frac{dV_c(t)}{dt} + V_c(t) = V$$

$$\mathcal{L}: RC\,(sV_c(s) - V_c(0)) + V_c(s) = V/s$$

$$\Rightarrow V_c(s) = \frac{V}{s(sRC+1)} = \frac{(V/RC)}{s(s+(1/RC))} = V\left\{\frac{1}{s} - \frac{1}{s+1/RC}\right\}$$

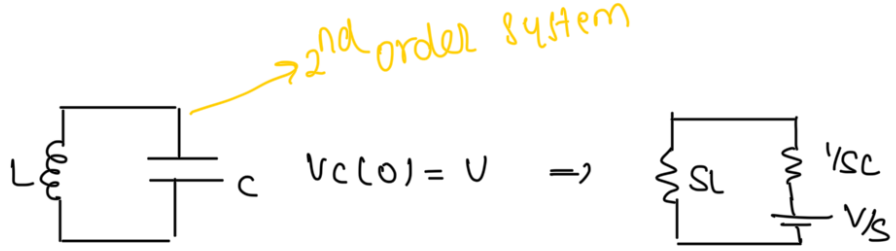
$$\Rightarrow v_c(t) = V\left[\mathcal{L}^{-1}(1/s) - \mathcal{L}^{-1}\left(1/(s+1/RC)\right)\right]$$

$$= V\,(1 - e^{-t/RC})$$

only resistors and fixed sources → always steady state

capacitors/Inductors | Time varying sources → changes

→ current/voltage source starts from rest ie steady state is attained as $t \to \infty$ from $-\infty$. Our interest is $t \ge 0$

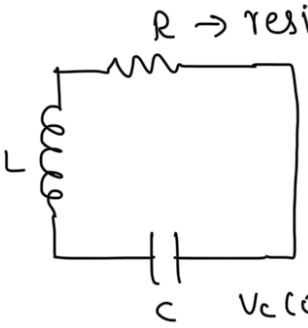


$$I(s) = \frac{V/s}{1/sC + sL} = \frac{Vc}{1 + s^2LC} = \frac{V/L}{s^2 + (1/LC)} = \frac{V/L}{s^2 - (j/\sqrt{LC})^2}$$

$$= \frac{V/L}{2j/\sqrt{LC}}\left\{\frac{1}{s - j/\sqrt{LC}} - \frac{1}{s + j/\sqrt{LC}}\right\} = -\frac{Vj}{2}\sqrt{\frac{C}{L}}\left\{\frac{1}{s - j/\sqrt{LC}} - \frac{1}{s + j/\sqrt{LC}}\right\}$$

$$I(t) = -Vj\sqrt{C}\left\{e^{jt/\sqrt{LC}} - e^{-jt/\sqrt{LC}}\right\}u(t) = -Vj\sqrt{C}\cdot 2j\sin(t/\sqrt{LC})\,u(t)$$

$$I(t) = \frac{V}{\sqrt{\frac{L}{C}}} \sin\left(\frac{t}{\sqrt{LC}}\right) u(t) \qquad \boxed{\omega = \sqrt{\frac{1}{LC}}}$$



$$I(s) = \frac{V}{s(sL + 1/sC + R)} = \frac{V}{s^2L + sR + 1/C}$$

$$= \frac{V/L}{s^2 + s(R/L) + 1/LC} = V/L \left[\frac{A}{s-r_1} + \frac{B}{s-r_2}\right]$$

$$r_1, r_2 = -\frac{RC \pm \sqrt{R^2C^2 - 4LC}}{2LC}$$

$(R/L)^2 > 4/LC$

$(R/2L)^2 > 1/LC$

$A_1 e^{r_1 t} + A_2 e^{r_2 t}$

(I) $(r_1, r_2 \rightarrow$ -ve & real$) \Rightarrow e^{r_1 t}, e^{r_2 t}$ + dying exponential

overdamped

$\Rightarrow$ No sinusoidal, only decay (Not RC decay)

$R^2C > 4L$

(II) $(r_1, r_2 \rightarrow$ imaginary$) \Rightarrow$ sinusoid terms are there

underdamped

$R^2C < 4L$

but since real terms are also there it'll be a decaying sinusoid

$r_1, r_2 = \sigma \pm j\omega_d$

$\sigma = \frac{R}{2L} \quad \omega_d = \sqrt{\frac{1}{LC} - \sigma^2} \Rightarrow i(t) = e^{-\sigma t}\{A e^{j\omega_d t} + B e^{-j\omega_d t}\}$

$e^{-\sigma t}$: decaying; $e^{\pm j\omega_d t}$: sinusoid

sinusoid with a dying amplitude

(III) $(r_1, r_2 \rightarrow$ equal$) \Rightarrow r_1, r_2 = -R$

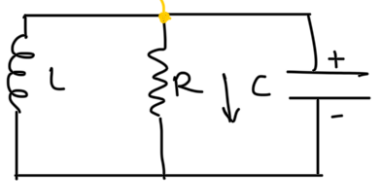
$\frac{}{2L}$

critically damped

$R^2C = 4L$

$\mathcal{L}^{-1}\left(\frac{1}{(S+R/2L)^2}\right) = te^{R/2L\,t}$

KCL here



$$\frac{V_c(S)}{R} + \frac{V_c(S)}{SL} + \frac{V_c(S) - V/S}{1/SC} = 0$$

$$V_c(S) = \frac{V \cdot C}{1/R + 1/SL + SC} = \frac{VS(LCR)}{SL + R + S^2LCR} = \frac{VS}{S^2 + S/RC + 1/LC}$$

$$= \frac{A}{S-r_1} + \frac{B}{S-r_2}$$

$$r_1, r_2 = \frac{-(1/RC) \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4\left(\frac{1}{LC}\right)}}{2}$$

(I) overdamped: $\left(\frac{1}{RC}\right)^2 > 4\left(\frac{1}{LC}\right) \Rightarrow L > 4R^2C$

(II) underdamped $\left(\frac{1}{RC}\right)^2 < 4\left(\frac{1}{LC}\right) \Rightarrow L < 4R^2C$

(decaying sinusoid)

(III) critically damped: $L = 4R^2C$

$\omega = \frac{1}{\sqrt{LC}}$ is called natural frequency/ undamped frequency

Type	condition	overdamped	critical	underdamped

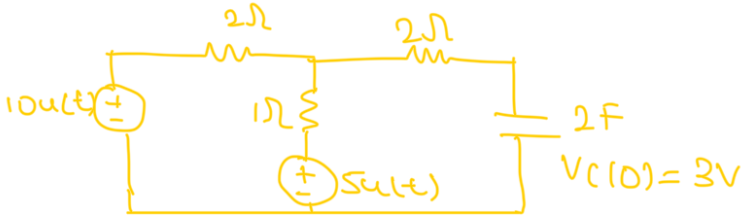
Series RLC	$\tau = L/R$	$\omega\tau < 1/2$	$\omega\tau = \frac{1}{2}$	$\omega\tau > \frac{1}{2}$
Parallel RLC	$\tau = RC$			

oscillation frequency of underdamped:

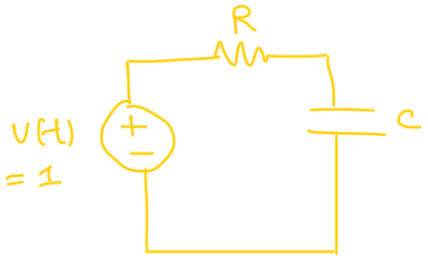
$$\omega_d^2 = \omega^2 - \left(\frac{1}{2\tau}\right)^2$$

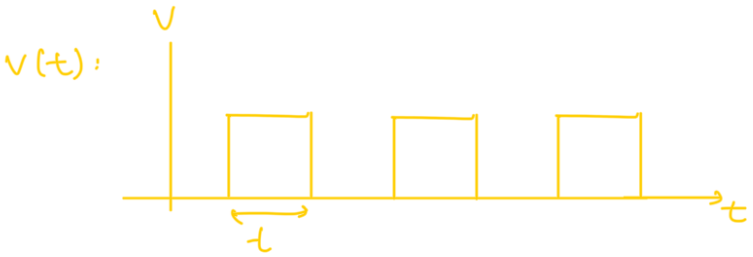
→ multiple sources ⟶ use superposition

Book uses $\frac{1}{2\tau}$ as α

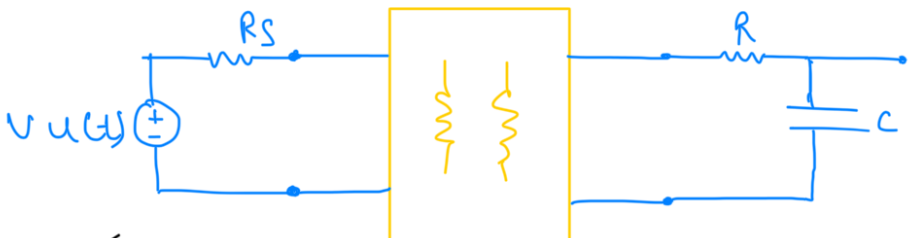


add 3 equations and source transformation





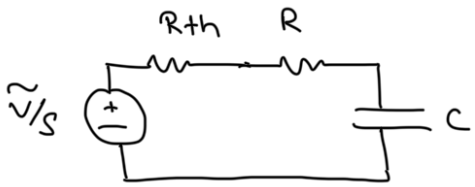
1% of V ⟹ $t = RC \ln 100$ $\quad$ $t = 0.99$ of V



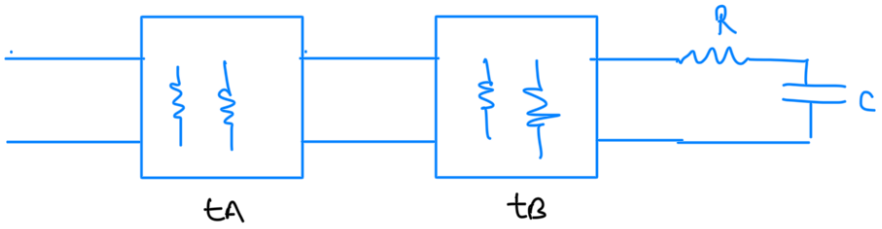
$v_c(t) = ?$

$v_c(0) = 0$

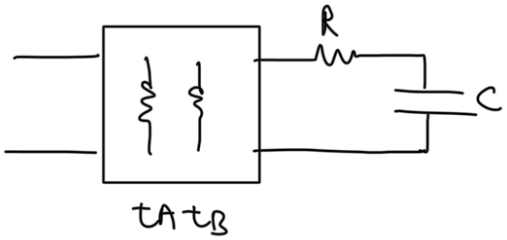
↳ If we get a therenin

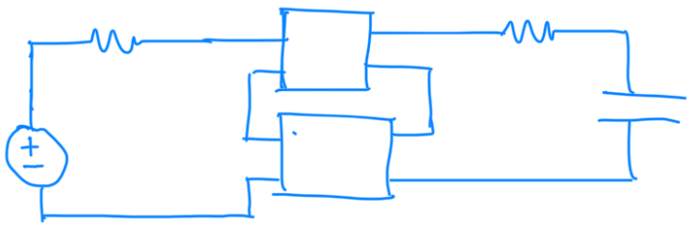


tutorial q



do in t-parameters ⇒





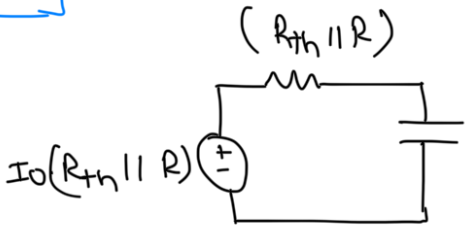
(series)

convert to Z ⇒ ZA + ZB

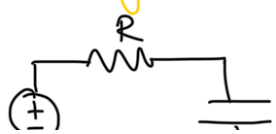
for parallel ⇒ convert to Y → Y = YA + YB



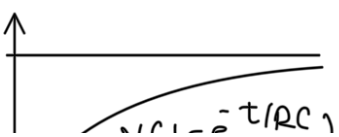
convert to Norton



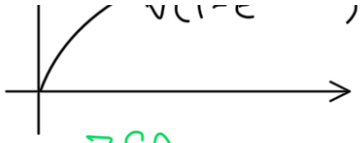
Terminologies:

① 

Vc(0) = 0

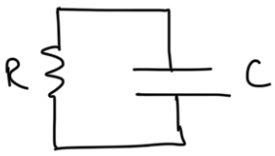




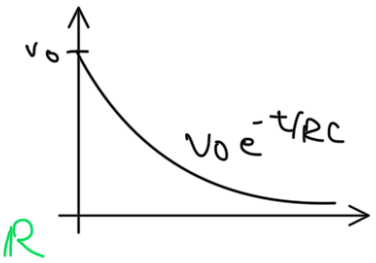


(zero state response) ZSR

(2)



$V_C(0) = V_0 \ (>0)$



(zero input response) ZIR

CDF inversion gen R.V

capacitor, var resis ⇒ give exp R.V

Differential equation view:

$$V_C(s) = \frac{A}{s^2 + as + b}$$

$$\Rightarrow (s^2 + as + b)\, V_C(s) = A$$

$$\Rightarrow \frac{d^2 v_c(t)}{dt^2} + a \cdot \frac{dv_c(t)}{dt} + b\, v_c(t) = c$$

$V_C(0) = 0$

$$\left.\frac{dv_c(t)}{dt}\right|_{t=0} = 0$$

{second order differential equation}

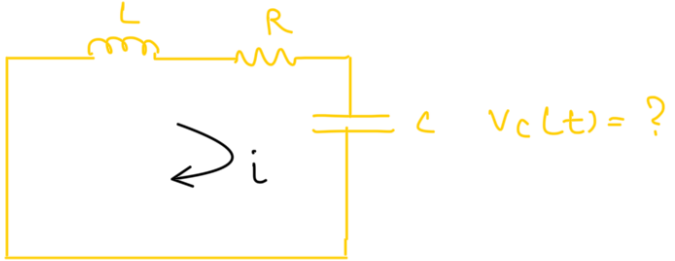
$$V_C(s) = \frac{A}{s+a}$$

$$\Rightarrow (s+a)\, V_C(t) = A$$

$$\Rightarrow \frac{d}{dt} v_c(t) + a\, v_c(t) = b$$

(denominator degree = order of diff.eqn)

{first order differential equation}



$$q(t) = C \cdot V_C(t)$$

$$i(t) = C \cdot \frac{dV_C(t)}{dt}$$

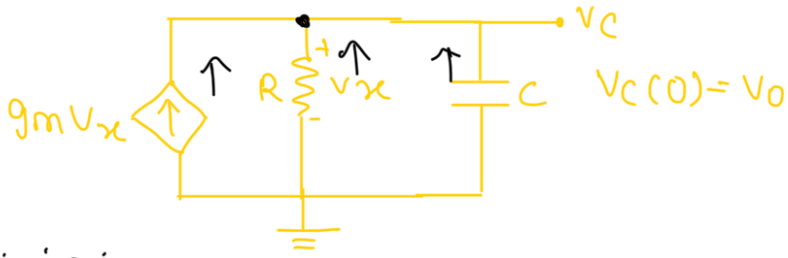
$$V_C(t) + i(t) \cdot R + L\frac{di(t)}{dt} = 0$$

$$\Rightarrow LC\frac{d^2V_c(t)}{dt} + RC\cdot\frac{dV_c(t)}{dt} + V_c(t) = 0$$

$$V_c(t) = \frac{1}{C}\int_0^t i(t)\,dt \;;\quad q(0) = 0$$

double derivative

$$\rightarrow V_L(t) = L\frac{di(t)}{dt}$$



Thevinin:

$$g_m V_x R = V_x \Rightarrow g_m R = 1 \quad \text{(no result)}$$

Nodal: $-\left(V_C(s) - \frac{V_0}{s}\right)sC - \frac{V_C(s)}{R} + g_m V_C(s) = 0$

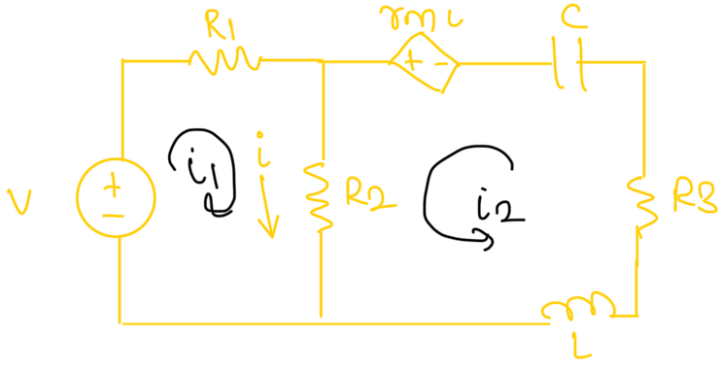
$$\Rightarrow V_C(s)\left[sC + 1/R - g_m\right] = V_0 \cdot C$$

$$\Rightarrow V_C(s) = \frac{V_0(RC)}{sRC - g_m R + 1} = \frac{V_0}{s + \left(\frac{1}{RC} - \frac{g_m}{C}\right)}$$

$$\Rightarrow V_C(s) = V_0\, e^{-(1/RC - g_m/C)t}$$

$g_m R > 1 \Rightarrow$ Voltage blows up (+ve feedback)

$g_m R < 1 \Rightarrow$ decays to zero eventually (-ve feedback)



Nodal / Mesh

$V - i_1 R_1 - (i_1 + i_2) R_2 = 0$

$$r_m(i_1+i_2) - (i_1+i_2)R_2 - i_2R_3 - SLi_2 - \frac{i_2}{SC} = 0$$

$$i_1(R_1+R_2) + i_2(R_2) - V = 0$$

$$i_1(r_m-R_2) + i_2\left(r_m-R_2-R_3-SL-\frac{1}{SC}\right) = 0$$

find i_2 and then $\frac{i_2}{SC} = V_C(S)$